AI-Assisted Literature Analysis for the Experimental Design of Solvent Optimization in Analgesic Studies of Plectranthus amboinicus

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Abstract (ACS)

This undergraduate research project examines how artificial intelligence (AI) can enhance literature-based experimental design in the search for bioactive and analgesic compounds from Plectranthus amboinicus (Lour.) Spreng. Using AI-driven text mining and systematic review methods, published studies were analyzed to identify solvent systems and extraction parameters most frequently associated with high yields of antioxidant and phenolic constituents linked to analgesic potential.

Computational clustering and bibliometric synthesis revealed consistent associations between solvent polarity and extract efficacy. Polar solvents—especially aqueous ethanol (20–40%) and methanol—were most often reported in studies showing high antioxidant or phenolic yield.

Research Context

- Natural product extraction and solvent optimization
- AI for literature synthesis and data mining
- Plectranthus amboinicus as a model for analgesic bioactivity

Objectives

1. Develop AI-assisted bibliometric pipeline.
2. Identify solvent systems maximizing extract yield.
3. Link polarity to reported analgesic efficacy.

Methodology Overview

AI-driven bibliometric workflow combining text mining, clustering, and structured data extraction.

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Development of the Bibliometric AI Assistant

A custom GPT-based agent was developed to:

- Extract solvent data from PDFs.
- Normalize terminology across studies.
- Generate structured datasets for analysis.

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Dataset Compilation

Dataset sourced from Scopus, PubMed, and Google Scholar.
Fields included: authors, year, solvent system, extraction yield, antioxidant/phenolic content.

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Extraction Parameters

Common extraction solvents:

- Methanol
- Ethanol (20–40%)
- Ethyl acetate

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Analytical Findings

Cluster analysis shows strong correlation between polarity index and yield.
High-performing solvents cluster near 0.7–0.9 dielectric constant.

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Analgesic Activity Correlation

AI extracted metrics of reported analgesic assays.
Solvent polarity aligns with increased total phenolic content and analgesic efficacy.

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Visualization Examples

- Heatmaps of solvent polarity vs. yield
- Network maps of co-occurring compounds
- Word clouds of major active terms

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Discussion

AI-assisted literature synthesis enables efficient exploration of extraction variables.
Potential extension to multi-objective optimization.

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Conclusions

- AI bibliometric tools streamline solvent analysis.
- P. amboinicus shows consistent results with polar solvents.
- Framework adaptable to other medicinal plant studies.

Acknowledgments

Supported by University of Puerto Rico – Río Piedras and ACS Puerto Rico Section.

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